

Talk 4. Modular Curves & Thm A

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Functor $F_{\Gamma, N} : \text{Sch}/\mathbb{Z}[1/N] \rightarrow \text{Set}$

$$S \mapsto \{EC/E/S \text{ w/ level } \Gamma(N)\}$$

$$\cdot \Gamma(N) : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow E[N]$$

$$\cdot \Gamma_1(N) : P \in E(S)[N] \text{ order } N$$

$$\cdot \Gamma_0(N) : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow E[N]$$

To be representable means $\exists Y_\Gamma(N) \rightarrow \text{Spec } \mathbb{Z}[1/N]$ w/

$$Y_\Gamma(N)(S) = \{EC \text{ w/ level } \Gamma(N)\}.$$

Taking $S = Y_\Gamma(N) \rightsquigarrow E \rightarrow Y_\Gamma(N)$ (univ.) w/ univ. level str. e.g. $\Gamma(N)$ (w.r.t. $\Gamma_1(N)$)

$$E \rightarrow E(N)$$

$$\downarrow \quad \downarrow$$

$$S \rightarrow Y(N)$$

Facts: For $N \geq 3$, $F_{\Gamma(N)}$ & $F_{\Gamma_1(N)}$ are rep. by a sim. aff. sch. / $\mathbb{Z}[1/N]$ called $Y(N)$, $Y_1(N)$, resp.

2) $F_{\Gamma_0(N)}$ rep. by a sim. DM stack $Y_0(N)/\mathbb{Z}[1/N]$, but NOT by a sch. bc. e.g. $[-1]$ is an automorphism. of order N

3) Can def. $Y_0(N)$ over $\mathbb{Z}$ (by replacing $\mathbb{Z}/N\mathbb{Z}$ w/ a fin. flat gp sch. "FFGS")

Compactify $Y_0(N) \rightsquigarrow X_0(N)$ since $Y_0(N)$ is not proper.

Def: A gen. ell. curve E/S is a proper flat sch. s.t.

1) E^{sm} has a gp str.

2) Fibres are ell. curves are n-gons

Def: An n-gon is $C_n := \mathbb{P}_R^1 \times \mathbb{Z}/n\mathbb{Z} / \sim$ where $(\infty, i) \sim (0, i+1)$

e.g. a 1-gon is a nodal curve $\bowtie$; In gen. $C_n^{sm} = \mathbb{G}_m \times \mathbb{Z}/n\mathbb{Z}$

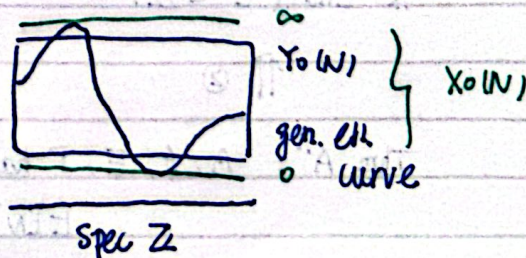
e.g. C_N has 2 level N -tors $\mu_N \subseteq \mathbb{G}_m, \mathbb{Z}/N\mathbb{Z}$

Add 2 pts (N prime):

$$- \infty := (C_1, \mu_N)$$

$$- 0 := (C_N, \mathbb{Z}/N\mathbb{Z})$$

In every fiber / $\mathbb{Z}$



e.g. $E/\mathbb{Q}$ everywhere semi-stab. redn.; $E_{\text{reg}}/\mathbb{Z}$ min. reg. model is a GEC because

- $E_{\text{reg}}^{\text{sm}} = E_{N\ell}^{\text{sm}}$ is a gp. sch.
- fibers are EC at good redn. (or generic fiber) are n -gon at multn. redn.

Thm A: $N > 7$ prime. Suppose $\exists$ ab var $A/\mathbb{Q}$ & $f: X_0(N) \rightarrow A$ st.

- 1) A has good redn away from N
- 2) $A(\mathbb{Q}) \text{ rk } 0$
- 3) $f(0) \neq f(\infty)$

Then $\# EC/\mathbb{Q}$ w/ $\mathbb{Q}$ -pt of order N .

$\Uparrow \textcircled{1}$

Thm A': A, f as above. Suppose $E/\mathbb{Q}$ w/ rat'l pt P of order N . Then

$$E[N] \simeq \mathbb{Z}/N\mathbb{Z} \oplus \mu_N$$

$\langle P \rangle$

"
 $\textcircled{1}$ Weil pairing $\langle, \rangle: E[N] \times E[N] \rightarrow \mu_N \xrightarrow{\sim} \langle P, - \rangle: E[N] \rightarrow \mu_N$
 $\Rightarrow \exists \text{ SES}$

has fin. fibers

$\downarrow$

$$0 \rightarrow \mathbb{Z}/N\mathbb{Z} \rightarrow E[N] \rightarrow \mu_N \rightarrow 0$$

2) f is non-const. $X_0(N) \rightarrow A + A(\mathbb{Q}) \text{ fin.} \Rightarrow X_0(N)(\mathbb{Q}) \text{ fin.}$

3) $\forall E/\mathbb{Q}, \text{End}_{\mathbb{Q}} E = \mathbb{Z}$ (for CM ell. curves, their extra end. not def.)
 $(\text{End}_{\mathbb{Q}} E \rightarrow \text{End}_{\mathbb{Q}}(T_0 E) = \mathbb{Q}) \text{ over } \mathbb{Q})$

Proof of $\textcircled{1}$: Suppose $P_1 \in E_1[N](\mathbb{Q})$. by $A' \Rightarrow E_1[N] = \mathbb{Z}/N\mathbb{Z} \oplus \mu_N$

$\Rightarrow$ set $E_2 = E_1/\mu_N, P_2 = P_1 \bmod \mu_N \Rightarrow \exists$ inf. chain

$$\begin{array}{ccccccc} E_1 & \rightarrow & E_2 & \rightarrow & E_3 & \rightarrow & \dots \\ \downarrow & & \downarrow & & \downarrow & & \\ P_1 & \mapsto & P_2 & \mapsto & P_3 & \mapsto & \dots \end{array}$$

$X_0(N)(\mathbb{Q}) \text{ fin.} \Rightarrow \exists i < j$ s.t. $(E_i, P_i) \xrightarrow{\sim} (E_j, P_j) \simeq (E, P)$. Now,

$E_i \rightarrow E_j$ is an endomorphism of E of deg N^{j-i} not killing P but
 $\ast \text{End}_{\mathbb{Q}} E = \mathbb{Z}$. $\square$

$\Uparrow \textcircled{2}$

Thm A'': A, f, E, P has above. p prime of bad redn for E . Then

$$E[N] \simeq \mathbb{Z}/N\mathbb{Z} \oplus \mu_N / \langle P \rangle$$

$\textcircled{2}$

②: $P: G_{\mathbb{Q}} \rightarrow \text{Gal}(\mathbb{F}_N)$ action on $E[N]$. From the SES,

$$P = \begin{pmatrix} 1 & f_0 \\ & \chi_N \end{pmatrix}$$

where f_0 is a 1-cocycle $\forall G_{\mathbb{Q}} \rightarrow \mathbb{F}_N$.

Let $K = \mathbb{Q}(\mu_N)$, then $P|_K = \begin{pmatrix} 1 & f \\ & 1 \end{pmatrix}$ w/ $f \in \text{Hom}(G_K, \mathbb{F}_N)$. WTS: $f = 0$.

PNP: Given Thm A'', $P|_K$ is everywhere unramified.

Pf: i) $p \neq N$ of good redn $\forall$ by NOS

ii) p bad redn. by A'' $\checkmark$

iii) $p = N$ good redn. $E = \text{Néron model} / \mathbb{Z}_p$ $E[N] = E[N] \otimes \times_{\mathbb{Z}_p} \mathbb{Q}_p$
 $0 \rightarrow \mathbb{Z}/N\mathbb{Z} \rightarrow E[N] \rightarrow \mu_N \rightarrow 0$

$\checkmark$ (which goes the opposite direction) $\uparrow$ FFGS

comm- $\bar{0}$ seq. $\Rightarrow E[N] = \mu_N \oplus \mathbb{Z}/N\mathbb{Z} \Rightarrow$ so is $E[N]$ □

$\Rightarrow f$ factors through the Hilbert cls field of K .

$\Rightarrow f \in \text{Hom}(G(K), \mathbb{F}_N)$

Under $\text{Gal}(K(\mathbb{Q}) \subset G(K)$, $f^\sigma = \chi(\sigma) f$

Herbrand's thm. $\Rightarrow f = 0 \Rightarrow f$ factors through the fin gp $\text{Gal}(K/\mathbb{Q})$.

Now, since $[K:\mathbb{Q}] = N-1$ coprime to N , P is semisimple. $\Rightarrow \checkmark$

Recall:

R a DVR, A ab var. / $\text{Frac}(R)$, then $\exists$ Néron model $\& A/R$ sm gp sch. NMP $\Rightarrow A(R) \hookrightarrow A(\text{Frac}(R))$

• Do the same w/ Dedekind domains

• $E/\mathbb{Q}$ $P \in E(\mathbb{Q})[N]$, $E/\mathbb{Z}$ Néron model

NOT P exists $\nexists P \in E(\mathbb{Z})$

E has add redn. if $E_{\mathbb{F}_p}^\circ \cong \begin{matrix} G_a, \mathbb{F}_p \\ G_m, \mathbb{F}_p \end{matrix}$ multi.

• Consequence of Raynaud's thm: $K/\mathbb{Q}_p$ e $\leq p-1$, G FFGS / $\mathbb{Q}_p$
 $G(\mathbb{Q}_p) \hookrightarrow G(K)$

Cor: A/K ab var. w/ good redn (or $(N, p) = 1$)

$$A[N](\mathbb{Q}_p) \hookrightarrow A[N](K).$$

Proof of Thm A''

semi-stab

• Step 1. Show E has everywhere redn.

③

• $p \neq N$. Suppose $E_{\mathbb{F}_p} = G_a \# E/E^0 \leq 4$ (also for special fibers of Néron models)
 Fact: $E_{\mathbb{F}_p}$ has ≤ 4 components $\Rightarrow P_{\mathbb{F}_p} \in E_{\mathbb{F}_p}^0(\mathbb{F}_p) = \mathbb{F}_p \Rightarrow P_{\mathbb{F}_p} = 0$
 $(N, p) = 1 \Rightarrow E[N](\mathbb{Z}) \hookrightarrow E[N](\mathbb{F}_p) \quad *$

• $p = N$. Suppose $E_{\mathbb{F}_p} = G_a$ Néron
 Fact: $\exists K/\mathbb{Q}_p$ deg ≤ 6 s.t. E/K has semi-stab redn. at p . $E'/\mathbb{Q}_K$
 $E'_K = G_{a,K}$; $E'_K{}^0 =$ ell. curve / torus

NMP: $f: E_{\mathbb{Q}_K} \rightarrow E' \rightsquigarrow \bar{f}: E_K \rightarrow E'_K$ $\bar{f}$ is triv. on $E'_K{}^0$. Now
 the argument $f(P)$ is the pt. extending P $E_{\mathbb{Q}_K} \rightarrow E' \xrightarrow{f} E'_K$ 4
 that $P_{\mathbb{F}_p} \in E_{\mathbb{F}_p}^0$ $P) \quad \downarrow \quad \downarrow \quad \downarrow$ since $\nexists$ non-const maps
 works when $p = N$ $P_K \in E'_K{}^0 \rightarrow E'_K{}^0 \quad 0$ from G_a to ell. cur.
 as well $\Rightarrow E'[N](\mathbb{Q}_K) \rightarrow E'[N](K)$ not inj. $*$ or torus $\square$

Lem: E has bad (mult.) redn. at $p = 2, 3$.

Pf: If not, Hasse b.d. $\# E(\mathbb{F}_p) \leq 7.5$ but $N > 7$. $\square$

(hence mult.)

Step 2. p prime bad redn. $P_{\mathbb{F}_p} \notin E_{\mathbb{F}_p}^0(\mathbb{F}_p)$

Pf: $\# E_{\mathbb{F}_p}^0(\mathbb{F}_p) = p-1$ or $p+1$ (split or non-split torus)

• $p = 2, 3$ $p+1 \leq N \quad \checkmark$

• $p = N$ $(p, p-1) = 1$, $(p, p+1) = 1 \quad \checkmark$

By Step 1, $\chi = (\nexists E, \mathbb{Z}/N\mathbb{Z} = \langle P \rangle) \in X_0(N)(\mathbb{Z})$:

$$0 = (C_N, \mathbb{Z}/N\mathbb{Z})$$

$$\infty = (C_1, \mu_N)$$

Else, $p \neq 2, 3, N$. Néron model $A/\mathbb{Z}$

$$A(\mathbb{Z})_{\text{tors}} \hookrightarrow A(\mathbb{Z}_p)_{\text{tors}}$$

$$\overset{\parallel}{A(\mathbb{Z})} \hookrightarrow \overset{?}{A(\mathbb{F}_p)} \overset{4}{\text{FFGS}} / \mathbb{Z}_p \text{ because } A \text{ has good redn. away from } N$$

E has mult red at 3. $P_{\mathbb{F}_3} \notin E_{\mathbb{F}_3}^0 \Rightarrow \underline{x \& 0 \text{ red to the same pt in } X_0(N)(\mathbb{F}_3)}$
 $\Rightarrow f(x), f(0) \in A(\mathbb{Z})$ red to the same pt of $A(\mathbb{F}_3) \Rightarrow f(x) = f(0)$. $\overset{\text{inj}}{\Rightarrow}$

If $P_{\mathbb{F}_p} \in E_{\mathbb{F}_p}^0$, then $x \& \infty$ red to the same pt. of $X_0(N)(\mathbb{F}_p)$. $\Rightarrow f(x) = f(\infty) \neq f(0) \neq f(\infty)$. $\square$

Cor: p bad redn. $E[N] = \mathbb{Z}/N\mathbb{Z} \oplus \mu_N$ (or $\mathbb{Q}_p$)

Pf: $E_{\mathbb{F}_p}^0 = \mathbb{G}_m$ has N -torsion $\mu_N \Rightarrow E_{\mathbb{F}_p}^0[N] \neq 0$ (order $\geq N$) but
 this doesn't intersect $\langle P \rangle = \mathbb{Z}/N\mathbb{Z} \Rightarrow E_{\mathbb{F}_p}[N] = E_{\mathbb{F}_p}^0[N] \oplus \mathbb{Z}/N\mathbb{Z}$
 $\Rightarrow E_{\mathbb{F}_p}[N] = \mu_N \oplus \mathbb{Z}/N\mathbb{Z}$. □